



DEALER BEST PRACTICE SERIES

3500 Engine Cylinder Head
Group Installation – Intake
Flange Alignment Gauge

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

3500 Engine Cylinder Head Group Installation – Intake Flange Alignment Gauge.....0

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1.0 Introduction

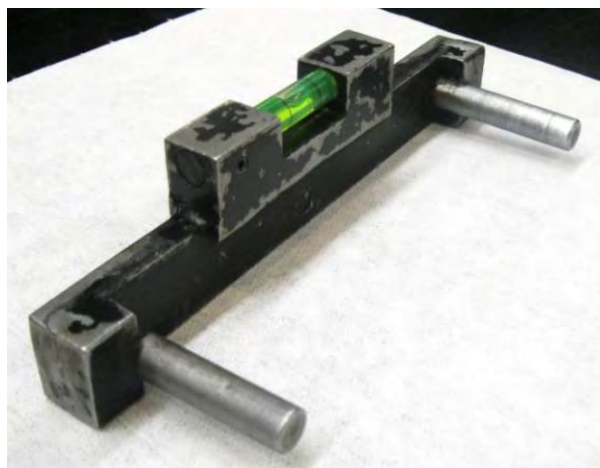
Many dealers today use a process to assemble and install the 3500 engine cylinder heads along with intake and exhaust manifolds as one complete group - the entire group on the block in one step. The process of installing the cylinder head group to the engine needs to be done with precision. Often, aligning the bolt's holes to the intake manifold with the aftercooler requires time and effort.



2.0 Best Practice Description

Ferreyros has developed a simple gauge to check the alignment of the intake flange bolt holes to assure that the flanges are mounted level at 180 degrees before installation begins. This check saves time spent adjusting the alignment while the group is installed on the engine block.

Since tool prints are not currently available, the following photos show the fixture and the process.



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Engine Cylinder Head Group Installation – Intake
Flange Alignment Gauge

DATE
2/19/2015

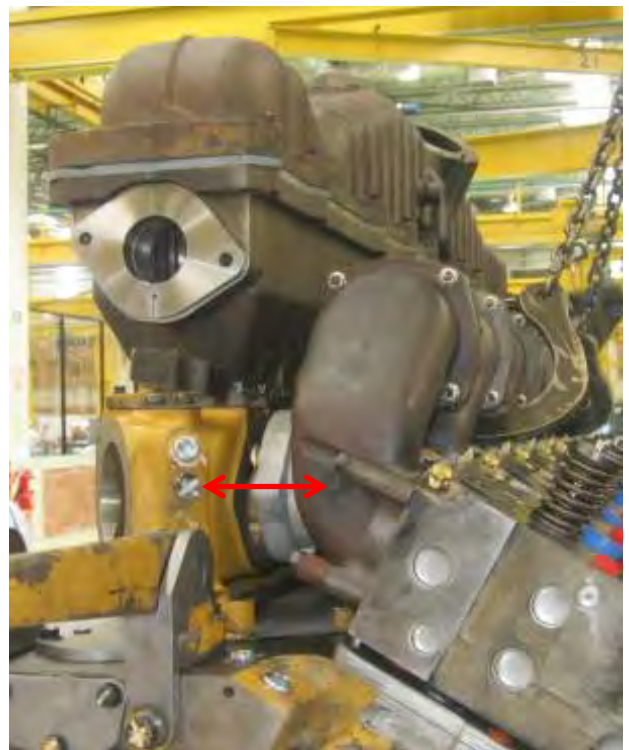
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Intake flange bolt holes are checked with the alignment gauge to make sure the flanges are mounted level. The check is performed while the cylinder head group is still mounted on the assembly stand.



The check has significantly reduced time spent adjusting alignment during cylinder head group installation.



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3.0 Implementation Steps

Design a tool similar to photos.

4.0 Benefits

- More efficient – reduce time of assembly.
- Reduce effort

5.0 Resources Required

Minimal expense to fabricate the tool

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

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8.0 Acknowledgements

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